OnePrompt-All: One Click to All Same-Concept Instances

A training-free point-conditioned grounding framework for SAM 3

Your Name · Your Institution · Supervisor: Name

Problem

Research question

Can one point trigger multi-instance segmentation in SAM 3?

OnePrompt-All asks whether a single point can act as a concept prompt. The click anchors SAM 3's pretrained visual representation and elicits multi-instance grounding without category text or additional training.

Interaction paradigm

PVS
Point → one instance

PCS
Text → concept set

OnePrompt-All
Point → concept set

Method

OnePrompt-All uses a positive click as a *geometric condition* rather than a category label. The click anchors the target in SAM 3's pretrained visual representation, while learned object queries search the full image in parallel for candidate instance masks. Lightweight post-processing then produces the final instance set without additional training.

Point-conditioned grounding

The click supplies sparse spatial evidence about the intended object. SAM 3 injects this cue into the grounding pathway, so the detector is conditioned by the selected location while still searching globally.

Candidate generation

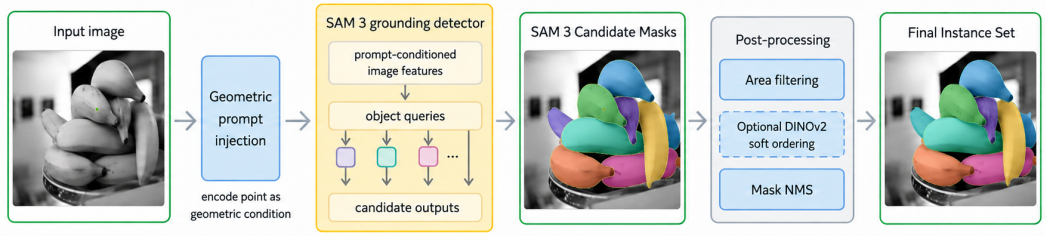
Multiple object queries attend to the conditioned image features in parallel. Their outputs form a high-recall set of scored masks that can include valid instances, weak fragments, and duplicate responses.

Filtering and deduplication

Area filtering first removes trivial fragments using a class-agnostic size rule. Mask NMS then suppresses highly overlapping predictions and keeps the stronger candidate, yielding independent instance masks.

Optional DINOv2 ordering

Before NMS, DINOv2 may softly reorder candidates by similarity to the clicked seed. It is auxiliary ranking rather than a hard semantic filter, so the core pipeline remains training-free and category-agnostic.



Results

Evaluation

The benchmark contains **2,220 images** from 232 classes and 11,100 trials. Each run uses RGB + one positive click; ground truth is used only for prompt sampling and scoring.

Main finding

OnePrompt-All reaches **F1 0.667** and **mIoU 0.601**, with precision 0.687 and recall 0.712. False candidates and click sensitivity remain the main errors.

Method	Benchmark	Images	Classes	Runs	F1	F1 SD	Prec.	Recall	mIoU	IoU SD
OnePrompt-All	Integrated OnePrompt-All Benchmark	2,220	232	11,100	**0.667**	0.280	0.687	0.712	**0.601**	0.279

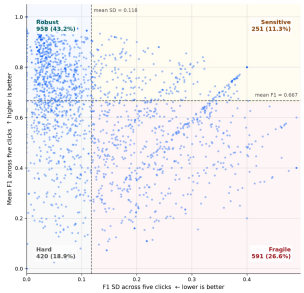


Fig. 2. Accuracy vs. click robustness across 2,220 images; dashed lines mark dataset means.

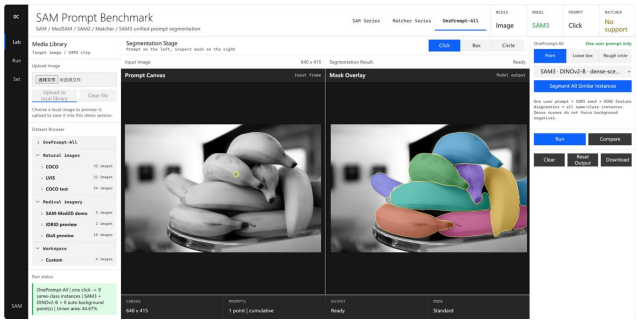


Fig. 3. Interactive workbench for prompting, model comparison, and mask inspection.

Conclusion & Discussion

Conclusion

OnePrompt-All is a **training-free** inference framework: a sparse instance-level click can condition SAM 3's pretrained visual representation and induce a useful multi-instance response.

Discussion

The key mechanism is one grounding detector with many learned object queries: multiple queries propose candidates in parallel, while area filtering and NMS clean fragments and duplicates.

Limitations

The click does not explicitly encode a category, so behavior remains sensitive to category, click location, occlusion, appearance variation, and distractors. Candidate calibration is the next focus.